



Time = 20 Minutes

Quiz No 6

[20 points]

Note - Provide all your answers on the answer sheet. Present your final answers and reasoning clearly. Most of the total score for the questions will be awarded on your reasoning. Please refrain from random or incoherent scribbles as they will not be graded.

Question 1 [20 points] Joint Distribution & Conditional Distribution with two Discrete Random Variables

You toss a biased coin two times, noting the value of the DRV X (X is the number Heads in the two throws). Therefore, X is a Binomial Random Variable with parameter $n = 2$ with probability of success (Heads) = 0.25.

(a) [2 points] Calculate and plot the P.M.F, $P_X(x)$ for the DRV X . (Hint – X can take on values 0, 1 & 2).

We now define a new DRV Y , where the conditional PMF $P(Y|X=x)$ is a discrete uniform distribution taking on values x to 3. (Therefore, if R.V $X = 0$, then the conditioned R.V $Y|X=0$ has a discrete Uniform distribution taking on values 0, 1, 2, 3. Similarly, if the R.V $X = 2$, then the conditioned R.V $Y|X=2$ has a discrete Uniform distribution taking on values 2, 3.)

(b) [4 points] Calculate and plot the conditional P.M.F $P_{Y|X}(y/x)$ for the DRV Y . (Hint – these will be three conditional pmfs $P_{Y|X}(y/x)$ for $X = 0, 1$ & 2).

(c) [8 points] Based on the working in Part (a) & (b), Calculate and plot the P.M.F, $P_Y(y)$ for the DRV Y .

(d) [6 points] Based on the working in Part (a), (b) & (c), Calculate and plot the Joint P.M.F, $P_{X,Y}(x,y)$ for the DRVs X and Y . (Hint – since R.V X takes on three values, 0, 1, 2 and R.V Y can take 4 values, $Y = 0, 1, 2, 3$, you may use a two-dimensional grid (see page 4) to write the joint probabilities.)

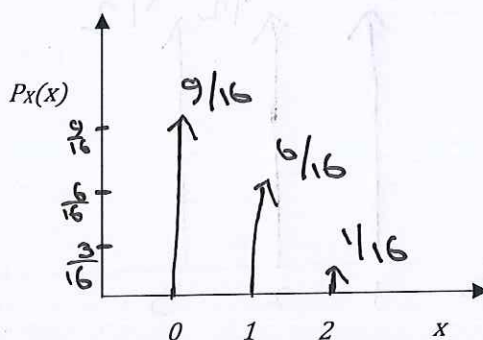
(a) [2 pts]

$X \rightarrow \text{Binomial}$
 $n=2, p=1/4$

$$P(X=0) = \binom{2}{0} \cdot \left(\frac{1}{4}\right)^0 \cdot \left(\frac{3}{4}\right)^2 = 9/16$$

$$P(X=1) = \binom{2}{1} \cdot \left(\frac{1}{4}\right)^1 \cdot \left(\frac{3}{4}\right)^1 = \frac{6}{16}$$

$$P(X=2) = \binom{2}{2} \cdot \left(\frac{1}{4}\right)^2 \cdot \left(\frac{3}{4}\right)^0 = \frac{1}{16}$$

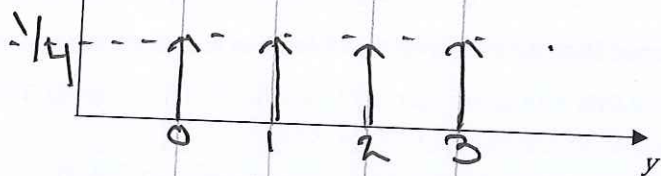


[1 pt]

[4pts]

(b)

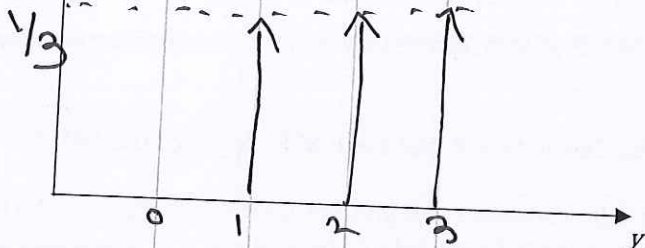
$P_{Y|X}(y|x=0)$



[1pt]

$\Rightarrow P_{Y|X}(y|x=0)$
is Uniform DRV
with $y=0, 1, 2, 3$

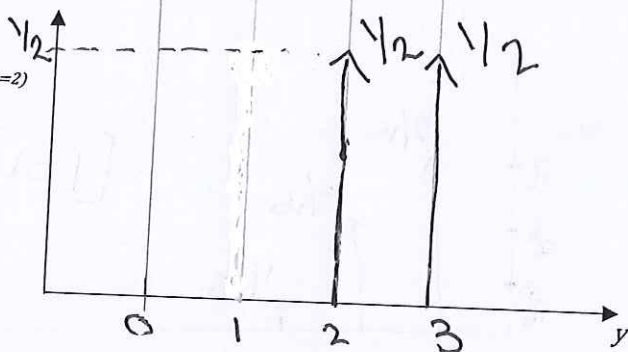
$P_{Y|X}(y|x=1)$



[1pt]

$\Rightarrow P_{Y|X}(y|x=1)$
is Uniform DRV
with $y=1, 2, 3$

$P_{Y|X}(y|x=2)$



$\Rightarrow P_{Y|X}(y|x=2)$
is Uniform DRV
with $y=2, 3$

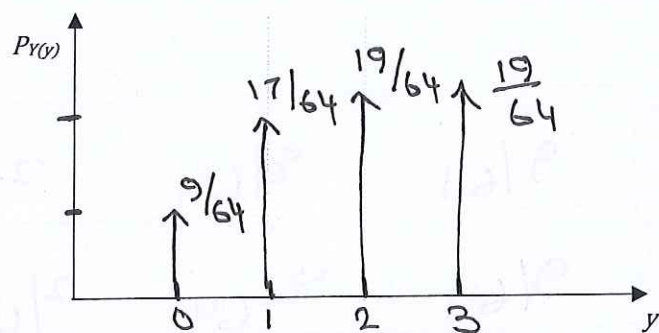
The Conditioned RV $Y|X$ is
Uniform from $y=x, \dots, 3$

$P_{Y|X}(y|x=x)$ is a uniform D.R.V
taking on values starting
from $x, \dots$ to 3 .

[1pt]

[8 pts]

(c)



Using Total Prob Theorem

$$P_Y(y) = \underbrace{P_{Y|X}(y|X=0) \cdot P_X(X=0)}_{(a)} + \underbrace{P_{Y|X}(y|X=1) \cdot P_X(X=1)}_{(b)} + \underbrace{P_{Y|X}(y|X=2) \cdot P_X(X=2)}_{(c)}$$

for $Y=y=0$

$$P_Y(0) = \underbrace{P_{Y|X}(0|X=0) \cdot P_X(X=0)}_{(a)} + \underbrace{P_{Y|X}(0|X=1) \cdot P_X(X=1)}_{(b)} + \underbrace{P_{Y|X}(0|X=2) \cdot P_X(X=2)}_{(c)}$$

$$\downarrow \quad \downarrow \quad \downarrow$$

$$\frac{1}{4} \cdot \frac{9}{16} + 0 \cdot \frac{6}{16} + 0$$

$$= \frac{9}{64} + 0 + 0 = \frac{9}{64}$$

for $Y=1$

$$P_Y(1) = P_{Y|X}(1|0) \cdot P_X(0) + P_{Y|X}(1|1) \cdot P_X(1) + P_{Y|X}(1|2) \cdot P_X(2)$$

$$= \frac{1}{4} \cdot \frac{9}{16} + \frac{1}{3} \cdot \frac{6}{16} + 0 = \frac{9}{64} + \frac{2}{16} = \frac{17}{64}$$

for $Y=2$

$$P_Y(2) = P_{Y|X}(2|0) \cdot P_X(0) + P_{Y|X}(2|1) \cdot P_X(1) + P_{Y|X}(2|2) \cdot P_X(2)$$

$$= \frac{1}{4} \cdot \frac{9}{16} + \frac{1}{3} \cdot \frac{6}{16} + \frac{1}{2} \cdot \frac{1}{16} = \frac{9}{64} + \frac{2}{16} + \frac{2}{64} = \frac{19}{64}$$

for $Y=3$

$$P_Y(3) = P_{Y|X}(3|0) \cdot P_X(0) + P_{Y|X}(3|1) \cdot P_X(1) + P_{Y|X}(3|2) \cdot P_X(2)$$

$$= \frac{1}{4} \cdot \frac{9}{16} + \frac{1}{3} \cdot \frac{6}{16} + \frac{1}{2} \cdot \frac{1}{16} = \frac{9}{64} + \frac{2}{16} + \frac{2}{64} = \frac{19}{64}$$

(d) Joint P.M.F, $P_{X,Y}(x,y)$ for the DRVs X and Y

6 pts

$Y=3$	$9/64$	$8/64$	$2/64$
$Y=2$	$9/64$	$8/64$	$2/64$
$Y=1$	$9/64$	$8/64$	0
$Y=0$	$9/64$	0	0
(X,Y)	$X=0$	$X=1$	$X=2$

$P_{X,Y}(x,y)$

[1 pt.]

$$P_{X,Y}(x,y) = P_X(x) \cdot P_{Y|X}(y|x)$$

[1 pt.]

$$P_{X,Y}(0,0) = P_X(0) \cdot P_{Y|X}(0|0) = \frac{9}{16} \times \frac{1}{4} = \frac{9}{64}$$

$$P_{X,Y}(0,1) = P_X(0) \cdot P_{Y|X}(1|0) = \frac{9}{16} \times \frac{1}{4} = \frac{9}{64}$$

$$P_{X,Y}(0,2) = P_X(0) \cdot P_{Y|X}(2|0) = \frac{9}{16} \times \frac{1}{4} = \frac{9}{64}$$

$$P_{X,Y}(0,3) = P_X(0) \cdot P_{Y|X}(3|0) = \frac{9}{64}$$

[4 pts]

Similarly $P_{X,Y}(1,0) = P_X(1) \cdot P_{Y|X}(0|1) = \frac{6}{16} \cdot 0 = 0$

$$P_{X,Y}(1,1) = P_X(1) \cdot P_{Y|X}(1|1) = \frac{6}{16} \cdot \frac{1}{3} = \frac{8}{64}$$

$$= P_{X,Y}(1,2) = P_{X,Y}(1,3) \quad (Y|X \text{ is Uniform b/w 1 to 3}).$$

Similarly $P_{X,Y}(2,0) = P_X(2) \cdot P_{Y|X}(0|2) = \frac{1}{16} \times 0 = 0$

$$P_{X,Y}(2,1) = P_X(2) \cdot P_{Y|X}(1|2) = \frac{1}{16} \times 0 = 0$$

$$P_{X,Y}(2,2) = P_X(2) \cdot P_{Y|X}(2|2) = \frac{1}{16} \times \frac{1}{2} = \frac{2}{64}$$

$$= P_{X,Y}(2,3) = \frac{2}{64}.$$